

Select and Pool Incidence Estimates

1. Select Available Incidence Results

Filter Dataset: Filter Endpoint Group: Filter: ☐ Groups Select study fields Check HIF Changes

Add to Pooling Window Add All

Endpoint	Endpoint Group	Dataset Name	Start Age	End Age	Author
☐ Acute Bronchitis	Acute Bronchitis	Additional Health Functions	8	12	Dockery et al.
☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Wheeze	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	17	Norris et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	99	Mar et al.
☐ Emergency Room Visits, Asthma	Emergency Room Visits, Respiratory	Additional Health Functions	0	99	Slaughter et al.

2. Select Pooling Methods

Remove Study Remove All Studies Pooling Window Name: Emergency Room Visits, Respiratory Pooling Levels: 0 Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Acute Bronchitis: Asthma Exacerbation: **Emergency Room Visits, Respiratory** x +

Studies, By Endpoint Pooling Method Nickname

Endpoint	Author	Qualifier	Study L...	Start A...	End Age	Age Ra...	Version	Year
Emergency Room Visits, R...	None	Emergency Room Visits, Respiratory	Emergency Room Visits, Asthma	Mar et al.; Slaug...	0	99	0-99	
Mar			Emergency Room Visits, Asthma	Mar et al.	0	99	0-99	1 2010
Slaughter			Emergency Room Visits, Asthma	Slaughter et al.	0	99	0-99	1 2005
Glad			Emergency Room Visits, Asthma	Glad et al.	0	99	0-99	1 2012

Preview Advanced Cancel Next

Each line in the pooling window represents a node in the tree structure, with each node representing either an individual incidence result or a collection of incidence results which have common values for their leftmost identifying variables. The tree structure is generated by comparing the leftmost values of the incidence result's identifying variables. High level nodes in the tree are formed when results have common values for identifying variables, and branches in the tree occur when the values differ.

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☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Wheeze	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.

2. Select Pooling Methods

Pooling Window Name: Pooling Levels: Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Acute Bronchitis: **Asthma Exacerbation** x Emergency Room Visits, Respiratory x | +

Studies, By Endpoint	Pooling Method	Nickname	Endpoint	Author	Qualifier	Study L...	Start A...	End Age	Age Ra...	Vers
☐ Asthma Exacerbation	None	Asthma Exacerbation	Asthma Exacerbation, Cough Asthma Exace...	Mar et al.; Ostro ...			6	18	6-18	
☐ Asthma Exacerbation, Cough	None	Asthma Exacerbation, Cough	Asthma Exacerbation, Cough	Mar et al.; Ostro ...			6	18	6-18	
☐ Mar			Asthma Exacerbation, Cough	Mar et al.	Uses in...	Spokan...	6	18	6-18	1
☐ Ostro			Asthma Exacerbation, Cough	Ostro et al.	Ostro st...	Los Ang...	6	18	6-18	1
☐ Asthma Exacerbation, Shortness of Breath		Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation, Shortness of Breath	Ostro et al.			6	18	6-18	
☐ Ostro			Asthma Exacerbation, Shortness of Breath	Ostro et al.	Ostro st...	Los Ang...	6	18	6-18	1
☐ Asthma Exacerbation, Wheeze		Asthma Exacerbation, Wheeze	Asthma Exacerbation, Wheeze	Ostro et al.			6	18	6-18	
☐ Ostro			Asthma Exacerbation, Wheeze	Ostro et al.	Ostro st...	Los Ang...	6	18	6-18	1

In the above example, four incidence results have been dragged into the pooling window. Each of the four health impact functions has **Endpoint Group** 'Asthma Exacerbation'. Thus, the top line, or root of the tree structure, represents all four incidence results. A branch then occurs in the tree structure, because two studies have **Endpoint** 'Asthma Exacerbation, Cough', while two others have **Endpoint** 'Asthma Exacerbation, Wheeze' and 'Endpoint Asthma Exacerbation, Shortness of Breath'. A further branch occurs within **Endpoint** 'Asthma Exacerbation, Cough' when **Author** of the two incidence results differs. Once a node has only a single incidence result, no further branching can occur.

Step 2. Sort results

After dragging incidence results into the pooling window, you can rearrange the order of the columns (variables), and thus change the tree structure. To do this, click on a column and hold the button down as you drag it to its new location. Note that the **Pooling Method** is always the first column after the **Studies, By Endpoint** column. All the other columns can be moved. To see how the order of the columns in the pooling window affects the tree structure, consider the following example:

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☐ Asthma Exacerbation, Cough	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Mar et al.
☐ Asthma Exacerbation, Wheeze	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.
☐ Asthma Exacerbation, Shortness of Breath	Asthma Exacerbation	Additional Health Functions	6	18	Ostro et al.

2. Select Pooling Methods

Pooling Window Name: Pooling Levels: Criteria used to group studies for pooling: P1 = Level 1 grouping; P2 = Level 2 grouping, etc.

Acute Bronchitis: **Asthma Exacerbation** x Emergency Room Visits, Respiratory x +

Studies, By Endpoint	Pooling Method	Nickname	Author	Endpoint	Qualifier	Study L...	Start A...	End Age	Age Ra...	Vers
☐ Asthma Exacerbation	None	Asthma Exacerbation	Mar et al.; Ostro...	Asthma Exacerbation, Cough Asthma Exace...			6	18	6-18	
☐ Mar et al.		Mar et al.	Mar et al.	Asthma Exacerbation, Cough			6	18	6-18	
☐ Mar			Mar et al.	Asthma Exacerbation, Cough	Uses in...	Spokan...	6	18	6-18	1
☐ Ostro et al.			Ostro et al.	Asthma Exacerbation, Cough Asthma Exace...			6	18	6-18	
☐ Ostro			Ostro et al.	Asthma Exacerbation, Cough	Ostro st...	Los Ang...	6	18	6-18	1
☐ Ostro			Ostro et al.	Asthma Exacerbation, Shortness of Breath	Ostro st...	Los Ang...	6	18	6-18	1
☐ Ostro			Ostro et al.	Asthma Exacerbation, Wheeze	Ostro st...	Los Ang...	6	18	6-18	1

This example uses exactly the same incidence results as the previous example, but with the **Author** column (variable) immediately after the **Pooling Method** and **Nickname** columns. Note that the **Pooling Levels** drop-down is set to 1 because one criterion (**Author**) is used to group studies for pooling. Multiple criteria (e.g., **Endpoint** and **Author**) may be used to group studies for pooling if desired.

BenMAP-CE Guidance and Best Practices

Once you have added health impact results to a pooling window, your pooling tree structure is determined by:

- (1) the number of **Pooling Levels** selected (up to 3), which determines the number of criteria used to group studies for pooling,
- (2) the order of the **Pooling Level** columns in the pooling window, and
- (3) the values of the relevant **Pooling Level** meta-data of the Health Impact Functions from which the incidence results were generated (**Endpoint Group**, **Endpoint**, etc.).

You may alter your pooling tree structure by modifying (1) using the drop-down menu and (2) by dragging and dropping the **Pooling Level** into the desired grouping hierarchy.

Step 3. Select pooling methods

Once the tree structure is set up in the pooling window, you are ready to select your pooling methods. Essentially each **Pooling Method** involves a different method of combining input incidence results to generate new incidence results. Results can be pooled any time a branch occurs in the tree structure — that is, any time two or more results share common values for their leftmost variables. BenMAP-CE helps you to identify these spots by inserting a value of *None* in the **Pooling Method** column at each spot where pooling is possible. These spots may change depending on the ordering of the columns and the selected number of **Pooling Levels**.

When a row that represents a group of incidence results is selected, an arrow appears at the right of the **Pooling Method** column, indicating the availability of a drop-down menu where you can select the desired **Pooling Method**. If you select **User Defined Weights** from the **Pooling Method** menu, a **Weight** column will appear to the right of the **Pooling Method** menu. All studies are assigned equal weight by default unless you update them. The sum of all **User Defined Weights** must equal 1.

If a **Pooling Method** is selected, you can view the calculated weights before executing the pooling by pressing the **Preview** button. In the case of the **Random or Fixed Effects** pooling method, the **Preview** button will reveal whether random or fixed effects are being applied.

Step 4. Create additional pooling windows if needed

Within a given pooling window, you can have only one ordering of the columns (variables). As we have seen, however, the ordering of the columns determines the structure of the tree used to pool results. It may thus sometimes be necessary for analyses to have multiple tree structures to handle the various pooling trees they require. To facilitate this, BenMAP allows additional pooling windows to be added and deleted. The pooling windows are displayed in a tabbed format.

To open a new pooling window, simply click on the rightmost tab marked with **+**. You may do this as many times as needed to accommodate different sort orders. You can add the same incidence results to as many different pooling windows as you like.

As needed, you can also delete a pooling window by clicking on the window you wish to delete and clicking the **x** to the right of the tab name.